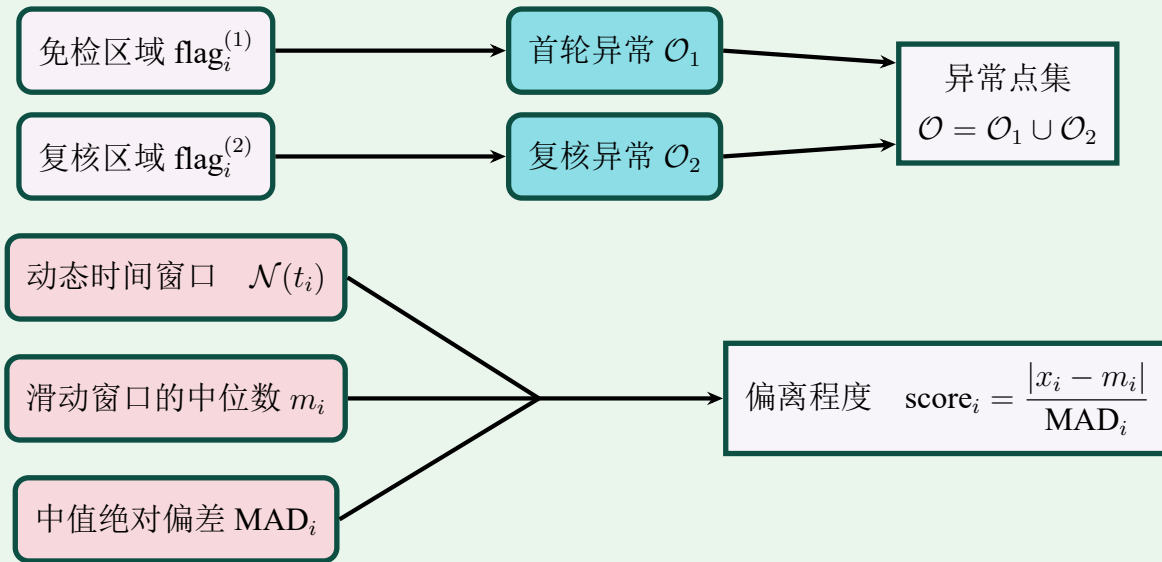
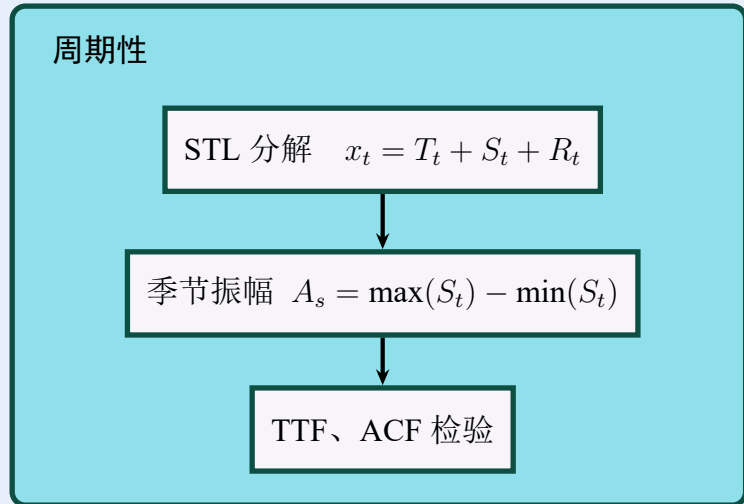


Step 1: 数据预处理
(Hample 滤波器)



Step 2: 分析数据特征
(STL 时间序列分解 FFT ACF)



Step 4: 透水率指标

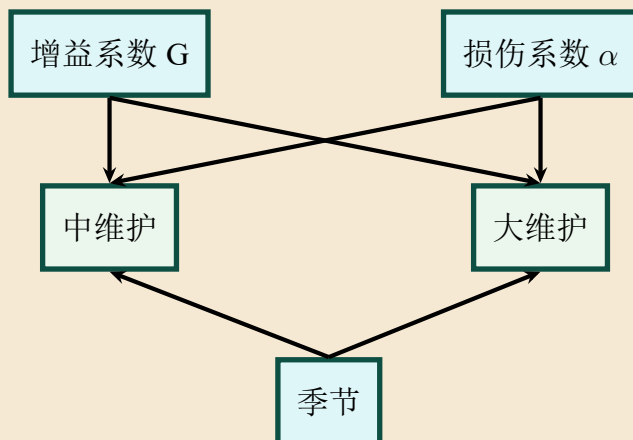
自然衰减 $k_0 = \frac{1}{M} \sum_{m=1}^M |k_m|$

季节影响 $\beta_s = \frac{\bar{k}_s}{k_0}, s \in \{\text{春, 夏, 秋, 冬}\}$

维护增益 $G = \frac{1}{n_a} \sum_{i=1}^{n_a} x(t_a^{(i)}) - \frac{1}{n_b} \sum_{i=1}^{n_b} x(t_b^{(i)})$

维护损伤 $\alpha = \frac{k'_i - k'_{i-1}}{k'_{i-1}}$

Step 3: 维护的影响



下降趋势

